



GEETANJALI DHOKE
business analyst

Functional Requirements Document (FRD)

GIS Data Visualization
& Analysis Tool

Project Title: GIS Data Visualization & Analysis Tool

Date:

Prepared By: Geetanjali Dhoke

Project Overview:

The GIS Data Visualization Tool will enable users to interact with spatial datasets for location intelligence, route planning, coverage analysis, and infrastructure monitoring. The tool will support map-based interactions, data filtering, and visual overlays of geographic features and attributes.

Business Requirements:

- ✓ Business users need an intuitive map interface to view and interact with spatial data.
- ✓ Analysts should be able to apply filters, visualize custom layers, and perform basic spatial analysis.
- ✓ Clients should receive location-based insights from multiple geospatial data sources.

User Roles & Permissions:

Role	Access Level
Viewer	Can view layers, interact with the map, and export visuals.
Analyst	Can filter, perform spatial queries, and draw custom shapes.
Admin	Can upload new datasets, manage user roles, and configure map layers.

Functional Requirements:

ID	Description	Priority	Source
FR-01	System shall fetch CSV/JSON files from Amazon S3 daily.	High	Business Team
FR-02	AWS Glue shall transform raw data based on business logic (e.g., date format standardization, null removal).	High	Data Team
FR-03	Transformed data shall be loaded into staging tables in Redshift.	High	Tech Lead
FR-04	A Lambda function shall validate the file structure and schema before ETL begins.	Medium	QA Analyst
FR-05	System shall log all job execution status and errors in CloudWatch.	High	DevOps Team
FR-06	SNS shall notify stakeholders via email upon success/failure of the pipeline.	Medium	Stakeholders
FR-07	Users shall be able to trigger manual runs via a web interface or API.	Low	Product Owner

Data Sources:

- ✓ Static GIS datasets (e.g., shapefiles, GeoJSON)
- ✓ Dynamic layers from APIs (e.g., traffic data, weather overlays)
- ✓ HERE platform or other real-time spatial data providers

System Architecture Overview:

- ✓ Frontend: Web-based map viewer (Leaflet.js, Mapbox, or OpenLayers)
- ✓ Backend: GIS server (GeoServer/PostGIS), APIs for data delivery
- ✓ Database: Spatially enabled DB like PostGIS
- ✓ Auth: Role-based login using OAuth or custom IAM
- ✓ Logs/Monitoring: CloudWatch or ELK Stack

Dependencies:

- ✓ Geospatial datasets and APIs
- ✓ Hosting platform (AWS, Azure, or on-prem)
- ✓ GIS services like PostGIS, GeoServer
- ✓ Web GIS libraries and frameworks